

AI CAREER COMPANION - PRODUCT KNOWLEDGE DOCUMENT

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Official System Specification, Architecture & User Guide

Version 1.0.0

1. Product Overview

AI Career Companion is an intelligent, end-to-end career guidance and candidate preparation platform. It bridges the gap between job seekers and employers by providing personalized recommendations, interview preparation, and document analysis.

- Product Name: AI Career Companion
- Main Purpose: Empower job seekers and students with AI-driven career recommendations, interview prep, and document analysis.
- Problem Solved: Eliminates manual skill-gap searching, generic interview prep, and lack of personalized resume alignment.
- Target Users: University students, software engineers, data scientists, and career transitioners.
- Key Objectives: Provide automated resume analysis, precision RAG internship matching, interactive mock interviews, and document Q&A.

2. Product Features and Functionalities

Feature 1: Automated Resume Parsing & Skill Extraction

Purpose: Extract candidate contact info, education, skills, and work experience from PDF/DOCX resumes.

How it works: PyMuPDF and python-docx parse file text, and regex NLP extract candidate metadata.

How users use it: Navigate to 'Resumes' tab and click 'Upload & Parse'.

Expected Result: Parsed structured profile with skills badge list.

Feature 2: Resume-Based Role Recommendation Engine

Purpose: Recommends optimal job titles, technical strengths, and suitable internship domains.

How it works: Match algorithm evaluates extracted candidate skills against technical role matrices.

How users use it: Click 'Role Recommendations' in sidebar and hit 'Generate Role Recommendations'.

Expected Result: Percentage fit scores, top technical skills list, and suitable internship titles.

Feature 3: AI Interview Preparation Agent

Purpose: Prepares candidates with role-specific technical questions, HR queries, and prep roadmaps.

How it works: AI agent structures role questions, STAR framework answer guidance, and 14-day study plans.

How users use it: Select target role under 'Interview Prep Agent' tab.

Expected Result: Complete Q&A guide, answer strategy, roadmap, and live interactive mock chat.

Feature 4: Document-Based Q&A System

Purpose: Extracts content from any PDF or DOCX file to answer questions and generate study QA pairs.

How it works: Extracts document text, stores context, and performs semantic search to answer queries.

How users use it: Upload document in 'Document Q&A' tab and enter questions in the chat box.

Expected Result: Summary, generated QA pairs, and specific answers based on document context.

Feature 5: RAG Internship Matcher

Purpose: Matches candidate profile against indexed internship database using FAISS vector search.

How it works: Candidate vector embedding is compared with FAISS indexed int